Utility Alerts — Stage 2 Go-Live Record

Thesis link: §5.4 limitation "Official utility alerts may arrive later than primary-source announcements"; §5.5 future work #3 "real-time official alerts". **Date recorded:** 2026-10-16 (FYP 2, Week 2) **Purpose:** close the loop on doc 04 §5 / doc 08 — the Stage 2 go/no-go decision was taken and executed on **2026-10-12**; this document records the live-delivery evidence accumulated since. All numbers queried live from production Supabase on 2026-10-16.

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1. Go-live decision

- **Decision:** flip `UTILITY_ALERT_SHADOW_MODE` from `true` to `false` (real delivery to opted-in users), owner-approved per the Stage 2 sign-off review in 08-utility-alerts-stage2-signoff (5/6 criteria pass; quiet-hours hold untested by data at the time).
- **Mechanism:** `UTILITY_ALERT_SHADOW_MODE = "false"` set in `[env]` of `fly.toml` (comment in file records the approval and date); no Fly secret overrides it — verified on the running machine with `printenv` (`false`).
- **First real delivery:** 2026-10-12 11:25 UTC — a few hours after the sign-off window closed, confirming the flip landed with the next deploy.

2. Live-delivery statistics (2026-10-12 → 2026-10-16, ~4 days)

Metric	Value
Real deliveries sent (`utility_alert_delivery` rows)	**934**
Distinct users who received ≥ 1 alert	52
Distinct alerts delivered	23
Alerts stored (detection side)	48
Users opted out (`wants_urgent_alerts = false`)	7 of 99
Deliveries per day	403 (10-12), 75, 162, 205, 89 (10-16, partial)

Reading the day-1 burst

The 403 deliveries on 2026-10-12 are the expected **go-live burst**: stored alerts whose *event time* still fell inside the relevance window (`UTILITY_ALERT_DELIVER_PAST_HOURS`) all became deliverable at once when shadow stopped swallowing them. This is the same burst class the 2026-10-12 "go-live burst fix" anticipated (relevance window keyed on `announced_at`, not storage time). Traffic then normalized to 75–205 deliveries/day.

3. Criterion closure vs doc 08

#	Criterion (doc 08)	Status after go-live
1	Precision ≥ 90%	[OK] held pre-go-live (100% on 20-alert sample); no user complaints recorded in the window as a negative signal
2	Zero pushes (shadow)	n/a — superseded by real delivery (this doc)
3	Quiet-hours hold 00:00–06:00 Kuching	still the only untested path; overnight Sarawak alert announcements remain rare
4	Rate cap ≤ 3/h per user	[OK] same code path live; day-1 burst respects the cap by construction
5	Opt-out wired end-to-end	[OK] verified in code 2026-10-16: `_matching_user_ids()` filters `wants_urgent_alerts IS TRUE` at SQL level; 7 users have opted out and are excluded from all delivery computation

4. What this evidence supports for the thesis

• **FYP2 future work #3 is now delivered, not just built** — official water/power disruption alerts are reaching real users (52 recipients in 4 days), with an opt-out mechanism (7 users) proving users can control it.

• The 48 stored / 23 delivered alert split shows the **location-matching + relevance-window filtering** working: roughly half of detections are either outside user locations or outside the delivery window — consistent with shadow-era precision.

• Remaining evaluation hooks for Phase 5: user response to alerts (retention of the toggle), and the first overnight announcement to close criterion 3.

5. Reproduction

```
# effective flag on the running machine
fly ssh console -a apanewsharitok -C "printenv UTILITY_ALERT_SHADOW_MODE"

# live counts (from the app machine, which holds DATABASE_URL)
fly ssh console -a apanewsharitok -C "python -c \"<sqlalchemy count queries>\""
```